

ARUN KARTHICK P

Software Developer | Python | SQL | Full-Stack Development | ETL Pipelines

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PROFESSIONAL SUMMARY

Computer Science Engineering student (3rd Year, CGPA 8.56) with hands-on experience building end-to-end software solutions — from ETL pipelines and predictive models to interactive dashboards and full-stack applications. 1st Place winner at the DND Data Arena 24-Hour Sprint, demonstrating the ability to deliver working, scalable solutions under pressure. Technical Manager at Data Forge, the official Data Science Club at SRMIST Trichy. Proficient in Python (Pandas, NumPy), SQL, and modern development tools; experienced in building ETL pipelines, predictive models, and automated workflows that turn requirements into reliable, well-tested software.

TECHNICAL SKILLS

Programming & Development: Python (Pandas, NumPy, Matplotlib, Seaborn), SQL, DAX, Power Query (M-Query)

Data & Reporting: SQL, Power BI, Tableau, Advanced Excel (VLOOKUP, PivotTables, Scenario Analysis), ETL Pipelines, Data Quality & Validation

Analytics & Engineering: Predictive Modelling, Data Pipeline Optimization, Workflow Automation, Cohort Analysis, Business Intelligence

Tools & Platforms: Streamlit, Plotly, DuckDB, Scikit-Learn, Apache Parquet, GitHub, MS Office Suite, Alteryx, RapidMiner

ACHIEVEMENTS & RECOGNITION

- 1st Place — DND Data Arena 24-Hour Sprint (Social Media Sentiment Track, Data n Dreads): Competed solo against teams in a rigorous 24-hour hackathon, delivering full-cycle data analysis — ingestion, modelling, dashboard structuring, and insight communication — within a strict deadline. Certificate issued by DND Data n Dreads.
- HackerRank SQL Gold Badge — Validates advanced proficiency in structured querying and relational data analysis.
- Organised a 24-Hour Hackathon at SRM Institute — Led end-to-end event coordination demonstrating planning, leadership, and cross-team collaboration.

LEADERSHIP & CLUB EXPERIENCE

Technical Manager | Data Forge — Data Science Club, SRMIST Trichy | 2024 – Present

- Lead technical initiatives, workshops, and project mentoring for the official Data Science Club at SRM Institute of Science & Technology, Tiruchirappalli.
- Drive end-to-end event management for data science events, hackathons, and competitive analytics activities for club members.

PROJECT EXPERIENCE

TATA IPL EcoDot Analytics | Python | Streamlit | Scikit-Learn | 2026

- Engineered an end-to-end Python ETL pipeline that ingested and processed 1,200+ raw ball-by-ball match JSON files containing 290k+ delivery rows to extract tournament-wide dot ball statistics.
- Implemented robust data cleansing and unification algorithms in Pandas to standardize historical team franchises and consolidated 20+ stadium name variations to ensure strict data integrity.
- Designed a dynamic multiplier mapping system to translate dot balls to tree plantation metrics, tracking 640k+ trees and calculating carbon offset potential based on IPCC environmental standards.
- Developed a Random Forest Regressor using Scikit-Learn to forecast future matchup dot-ball counts and environmental contributions based on venue, batting team, and seasonal factors.
- Conceptualized a proprietary "Green Impact Score" metric to objectively rank the most eco-efficient restrictors and bowlers across historical seasons.
- Built a premium dark-themed glassmorphic dashboard in Streamlit using Plotly and custom CSS with responsive micro-animations and smooth transitions.

SYNAPSE — Synthetic Data Intelligence Platform | Python | DuckDB | Streamlit | Scikit-Learn | 2025

- Architected an end-to-end synthetic data pipeline that simulates real-world datasets across domains (ecommerce, IoT, fintech) and processes data using a layered event → silver → gold architecture with DuckDB.
- Integrated CTGAN-based synthetic data generation with a fallback engine for stability; validated data via distribution similarity, correlation fidelity, and ML utility (train-on-real/test-on-synthetic cross-validation).
- Engineered a composite Trust Score metric combining multiple validation dimensions into a single interpretable confidence index; generated datasets up to 50,000+ rows with pipeline runtime optimized to under 2 minutes.
- Achieved ML utility scores of ~0.7; built an interactive Streamlit dashboard with real-time insights and dataset download capability, resolving deployment challenges across local and cloud environments.

Brand Pulse Monitor (SentimentScope) — DND Winning Dashboard | *HTML5 | Chart.js | PapaParse | Vanilla JS | 2024*

- Delivered a clarity-first HTML/CSS/JS dashboard analysing 732 social media posts across Twitter, Instagram, and Facebook covering 10+ countries — winning 1st Place at DND Data Arena 24-Hour Sprint.
- Implemented in-browser CSV parsing (PapaParse), dynamic filter logic (platform/country/month), and 8 interactive chart types (Bar, Donut, Line, Gauge, Heatmap, Stacked Bar) using Chart.js 4.4.1.
- Designed a live Auto Story Headline feature that auto-narrates the full data story and updates with every filter change — no backend, no frameworks, zero installation required.

Freelancer Platform Funnel & Growth Paradox Analysis | *SQL | PostgreSQL | Power BI | 2025*

- Diagnosed the 'Growth Paradox' — why marketplace scale doesn't always translate into revenue efficiency — by building an end-to-end analytics project using PostgreSQL and Power BI.
- Analysed revenue efficiency vs. growth trends, category-level supply-demand imbalance, client & freelancer funnel conversion leakage, and revenue concentration risk across 35,000+ interaction records.
- Built scalable cohort-driven ETL workflows and an interactive Power BI decision dashboard to support strategic growth optimization scenarios for non-technical business stakeholders.

Digital Wellbeing AI Dashboard | *Python | Streamlit | AI-Powered | 2025*

- Built an AI-powered analytics dashboard to analyze and visualize digital wellbeing patterns from user behaviour data, surfacing actionable insights on screen time, app usage, and focus habits.
- Deployed interactive Streamlit interface with real-time visualizations; available on GitHub.

INDUSTRY EXPERIENCE (SIMULATION PROGRAMS)

- Tata (Forage) — GenAI Powered Data Analytics Virtual Experience
- Deloitte (Forage) — Data Analytics Virtual Experience
- EduSkills — Google AI-ML Virtual Internship

CERTIFICATIONS

- Microsoft DP-600 (Fabric Analytics Engineer Associate) — 907/1000
- Google Advanced Data Analytics Certificate
- HackerRank Advanced SQL Gold Badge
- Microsoft ETL Fundamentals Certification

EDUCATION

B.Tech Computer Science & Engineering | SRM Institute of Science & Technology, Tiruchirappalli | *Jun 2024 – Jul 2028*
CGPA: 8.56 • 3rd Year (Class of 2028)